

Project Design Phase Solution Architecture

Date	14 February 2026
Team ID	LTVIP2026TMIDS82993
Project Name	HouseHunt: Finding Your Perfect Rental Home
Maximum Marks	4 Marks

Solution Architecture:

The HouseHunt system is designed using a **3-Tier Architecture Model**, which bridges the gap between business problems (difficulty in rental property management) and technology solutions (a secure MERN-based web application).

The architecture ensures:

- Secure authentication
- Clear separation of concerns
- Scalable design
- Efficient data handling
- Maintainable modular code structure

Architecture Type: 3-Tier Web Application Architecture

The system consists of three primary layers:

1 Presentation Layer (Frontend)

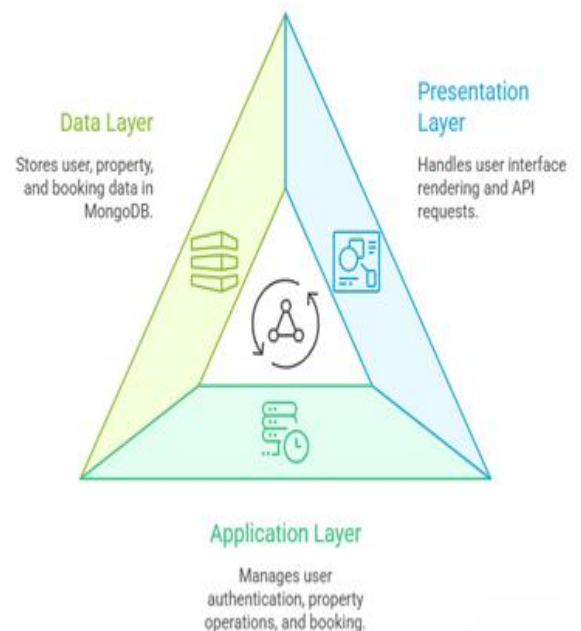
Technology Used:

- React.js
- HTML5
- CSS3
- JavaScript
- Axios

Responsibilities:

- User Interface rendering
- Form handling (Login, Register, Add Property)
- Dashboard display (Owner / Renter / Admin)
- Sending API requests to backend

3-Tier Web Application Architecture



- Displaying property listings and booking information

User Types:

- Admin
- Owner
- Renter

2 Application Layer (Backend)

Technology Used:

- Node.js
- Express.js
- JWT
- bcrypt.js
- REST APIs

Responsibilities:

- User authentication & authorization
- Role-based access control
- Property CRUD operations
- Booking management
- Middleware validation
- Error handling

Backend Modules:

- User Controller
- Property Controller
- Booking Controller
- Auth Middleware
- Role Middleware

3 Data Layer (Database)

Technology Used:

- MongoDB
- Mongoose ODM

Responsibilities:

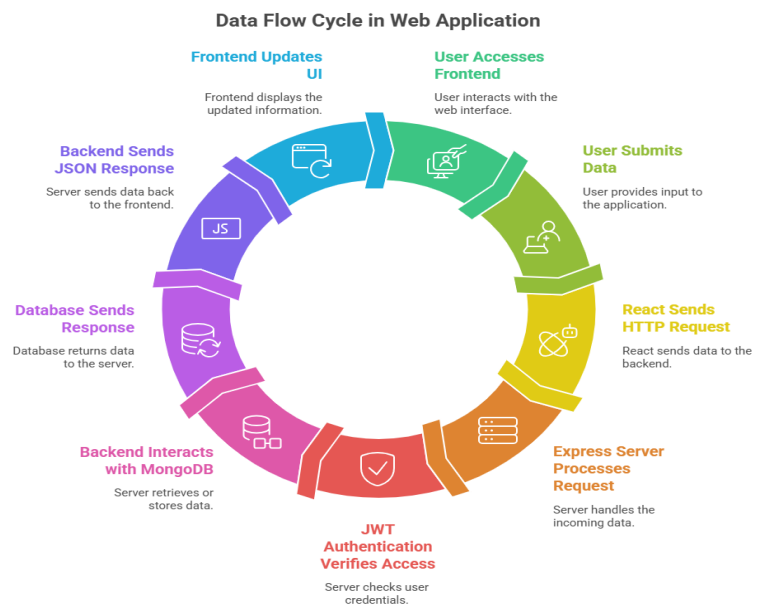
- Store user details
- Store property details
- Store booking records
- Maintain data relationships

Collections:

- Users
- Properties
- Bookings

Data Flow Architecture

1. User accesses frontend application.
2. User submits data (login, registration, property details).
3. React sends HTTP request to backend using Axios.
4. Express server processes request.
5. JWT authentication verifies access.
6. Backend interacts with MongoDB.
7. Database sends response back to backend.
8. Backend sends JSON response to frontend.
9. Frontend updates UI accordingly.



Security Architecture

- Password hashing using bcrypt
- JWT-based authentication
- Role-based authorization
- Protected API routes
- CORS enabled
- Environment variable configuration